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Different Proofs

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Chapter 1

Fermat's Little Theorem

1.1 Statement

1.2 Proof using Binomial theorem

Theorem 1 (Fermat's Little Theorem using the Binomial Theorem)**.** *$FermatLittleTheorem_Binomial For any pr$*
$a \pmod{p}$.

Proof. Use Freshmen's dream: $(a+b)^p = a^p + b^p$ in characteristic p. Then $(a+1)^p \equiv a^p + 1 \pmod{p}$, and the result follows by induction on a. □

1.3 Proof using Lagrange's theorem

Theorem 2 (Fermat's Little Theorem using the Lagrange's theorem)**.** *$FermatLittleTheorem_Lagrange For any p$*
$a \pmod{p}$.

Proof. If a is divisible by p, then $a^p \equiv 0 \equiv a \pmod{p}$. Assume that a is not divisible by p. Consider the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ of nonzero elements modulo a prime p. This group has order $p-1$. By Lagrange's theorem, the order of any element divides the order of the group. Therefore, for any integer a not divisible by p, we have $a^{p-1} \equiv 1 \pmod{p}$. □

1.4 Proof using Dynamical Systems

Definition 3 (Definition of T_n)**.** T For a natural number n (≥ 2), define $T_n : [0,1] \to [0,1]$ by $T_n(x) = \{nx\}$ for $0 \leq x < 1$ and $T_n(1) = 1$, where $\{y\}$ denotes the fractional part of y.

Lemma 4 (Composition law for T_n)**.** *def:T* $T_comp_eq_m ul For any natural numbers m,$
$n, the composition of\ T_m$ *and* T_n *equals* T_{mn}*, i.e.,* $T_m \circ T_n = T_{mn}$.

Proof. If $x = 1$, then both sides equal 1. Otherwise $T_n(x) = \{nx\} \in [0, 1)$, and $T_m(T_n(x)) = \{m\{nx\}\}$. Writing $m\{nx\} = mnx - m\lfloor nx \rfloor$ and noting that $m\lfloor nx \rfloor \in \mathbb{Z}$, we have $\{m\{nx\}\} = \{mnx\} = T_{mn}(x)$. □

Lemma 5 (Number of fixed points of T_n)**.** *def:T* $T_num_fp_eqForanyn \geq 2$*, the map* T_n *has exactly* n *fixed points.*

Proof. For $x \in [0, 1)$, the equation $T_n(x) = x$ becomes $\{nx\} = x$, i.e., $(n-1)x \in \mathbb{Z}$. With $x \in [0, 1)$, this gives $x = j/(n-1)$ for $j \in \{0, 1, \ldots, n-2\}$, contributing $n - 1$ fixed points. Together with the fixed point $x = 1$, this gives exactly n fixed points. □

Theorem 6 (Fermat's Little Theorem using Dynamical Systems)**.** $FermatLittleTheorem_DynamicalForanyprin$
$a \pmod{p}$*.*

Proof. lem:T-num-fp-eq,lem:T-comp-eq-mul,def:T By the standard reduction to the natural-number version, it suffices to prove the statement for natural numbers. The cases $a = 0$ and $a = 1$ are trivial; assume $a \geq 2$. Let S be the set of fixed points of T_{a^p} that are not fixed points of T_a. By **??** applied to T_{a^p} and T_a, S has $a^p - a$ elements. For each $x_0 \in S$, since x_0 is fixed by $T_{a^p} = T_a^p$ (using **??** iteratively), the T_a-orbit of x_0 has size dividing p. Because p is prime, the size is 1 or p; size 1 would make x_0 a fixed point of T_a, which is excluded. Hence every orbit has exactly p elements, and these orbits partition S. Therefore p divides $|S| = a^p - a$. □